

Behavioral biases of cryptocurrency investors: a prospect theory model to explain cryptocurrency returns

Behavioral
biases of CC
investors

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Abstract

Purpose – The study aims to test prospect theory (PT) predictions in the cryptocurrency (CC) market. It proposes a new asset pricing model that explores the potential of prospect theory value (PTV) as a significant predictor of CC returns.

Design/methodology/approach – The study comprehensively analyses a large sample set of 1,629 CCs, representing more than 95% of the CC market. The study uses a portfolio analysis approach, employing univariate and bivariate sorting techniques with equal-weighted and value-weighted portfolios. The study also employs ordinary least squares (OLS) regression, panel data methods and quantile regression (QR) to estimate the models.

Findings – This study demonstrates an average inverse relationship between PTV and CC returns. However, this relationship exhibits asymmetry across different quantiles, indicating that investor reactions vary based on market conditions. Moreover, PTV provides more robust predictions for smaller CCs characterized by high volatility and illiquidity. Notably, the findings highlight the dominant role of the probability weighting (PW) component in PT for predicting CC behaviors, suggesting a preference for lottery-like characteristics among CC investors.

Originality/value – The study is one of the early studies on CC price dynamics from the PT perspective. The study is the first to apply a QR approach to analyze the cross-section of CCs using a PT-based asset pricing model. The results shed light on CC investors' decision-making processes and risk perception, offering valuable insights to regulators, policymakers and market participants. From a practical perspective, a trading strategy centered around the PTV effect can be implemented.

Keywords Asset pricing model, Behavioral biases, Cryptocurrency, Prospect theory, Quantile regression

Paper type Research paper

1. Introduction

How individuals evaluate risk and make investment decisions in risky and uncertain situations has always been the center point of debate in the financial markets literature (Barberis *et al.*, 2016; Tversky and Kahneman, 1992). Conventionally, academic literature has often employed expected utility theory (EUT) and models of rational equilibrium with adjustments for behavioral biases to model this individual investment decision-making process. However, in modern electronic markets with a plethora of investment choices (e.g. cryptocurrencies or “CCs”) and voluminous trading activity by retail investors, considerable evidence has become available that investors do not conform to EUT and models of rational equilibrium (Barberis *et al.*, 2021; Chen *et al.*, 2022). Therefore, predictions of EUT about investors' attitudes towards risk and investment decision-making are being increasingly



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challenged (Bondt and Thaler, 1985; Tversky and Kahneman, 1974, 1981, 1992). More recently, an alternative construct of individual decision-making, namely the prospect theory (PT) model of choice, has been adopted from the area of behavioral psychology (Kahneman and Tversky, 1979; Tversky and Kahneman, 1992). In the behavioral finance literature, PT has emerged as the popular alternative to EUT as it accurately explains investors' decision-making and attitude toward risk from real-life observed behavior (Barberis *et al.*, 2016, 2021). It has been successfully applied across various asset classes- in bonds (Bondt and Thaler, 1985), in stocks (Barberis *et al.*, 2016), in the foreign exchange market (Kozhan *et al.*, 2020), and in foreign share markets (Wang *et al.*, 2021), to explain the decision-making by investors. Specifically, PT has been used to understand and account for various phenomena in finance, like the equity premium puzzle (Barberis and Huang, 2008; Benartzi and Thaler, 1995), momentum (Grinblatt and Han, 2005) and disposition effect (Barber and Odean, 2008; Barberis and Xiong, 2009; Hens and Vlcek, 2011; Li and Yang, 2013).

Moreover, it seeks to explain how certain behaviors like loss aversion, risk-taking and preference for lottery-like payoffs lead investors to make irrational decisions that result in several market anomalies (Bali *et al.*, 2011; Barberis *et al.*, 2016; Benartzi and Thaler, 1995; Boyer *et al.*, 2010; Harvey and Siddique, 2000). Applying the PT in the context of stock returns, Barberis *et al.* (2016) simplified the decision-making process into two stages. In the first stage, investors create a mental representation of the historical return distribution, as they are the best proxy for estimating future returns. In the second stage, they evaluate these distributions in the way described by PT. In this study, I propose an asset pricing model involving investors taking each security's historical return distribution and computing the prospect theory value (PTV). Securities with high PTV are more appealing and more likely to be part of the portfolio, while those with low PTV are less appealing and more likely to be excluded. I aim to test the assumption that high PTV security will earn lower subsequent returns. One possible explanation Barberis *et al.* (2016) suggested is the overvaluation hypothesis. Securities with high PTV are attractive to investors, leading to overvaluation and ultimately earning lower future returns. I aim to provide empirical evidence for the impact of PTV on asset prices.

Against this backdrop, CC markets provide a fertile ground fittingly apt to test the PT model (Chen *et al.*, 2022). This is so because CCs lack fundamental economic value (Cheah and Fry, 2015; Mnif *et al.*, 2022); therefore, investor behavior is a defining component of the CC market. These CC markets are characterized by (1) a lack of fundamental value, (2) the absence of regulatory constraints, (3) the absence of support from central agencies as in the case of legal tender fiat currency, and (4) a global retail investor profile that is driven by sentiments [1] (Corbet *et al.*, 2019; Subramaniam and Chakraborty, 2019).

Moreover, the CC markets have witnessed tremendous growth over the last decade. They have attracted considerable attention from all the market stakeholders, including the investor community, popular press, regulators and policymakers, governments and academics. The number of CCs has increased exponentially since the introduction of Bitcoin in 2009 to more than 11,000 today, with a current market capitalization of over USD 2 trillion (January 2023) [2]. With rising interest in CC markets, academic literature has also flourished recently. The most prominent strands of CC literature include (1) informational efficiency (Tiwari *et al.*, 2018; Urquhart, 2016), (2) liquidity and price discovery (Tripathi *et al.*, 2022), (3) bubble formation (Cheah and Fry, 2015), (4) arbitrage opportunities (Makarov and Schoar, 2020) (5) speculative and lottery demands (Cheah and Fry, 2015; Grobys and Junttila, 2021) (6) hedging and diversification benefits of CCs (Corbet *et al.*, 2019), among others. The extant literature predominantly employs traditional empirical asset pricing models (e.g. 3-factor, 5-factor models) (Fama and French, 1993, 2015) to explain the cross-sectional of CC returns. However, there is ample prior evidence from behavioral finance literature to show that these models do not adequately explain the pricing dynamics of real financial markets, and even more so of

CC-like instruments (Chen *et al.*, 2022; Grobys and Junttila, 2021). I aim to fill this research gap by applying the principles of PT to augment the extant asset pricing approaches and thus better understand the price dynamics of CCs and provide improved explanations for investor decision-making in CC markets. In this study, I will utilize data from 1,629 CCs over nine years (2014–2022) obtained from coinmarketcap.com. These currencies represent over 95% of the entire CC market in terms of market capitalization value, ensuring that my findings have broader applicability and generalizability.

My hypothesis receives empirical support through both portfolio-level analysis and cross-section regressions. I construct the decile portfolios based on the PTV on $t-1$ day and rebalance it on daily frequency to justify speedy trading in the CC market. The average raw and risk-adjusted return (Capital Asset Pricing Model [CAPM] and three-factor) difference between CCs in the lowest and highest PTV deciles is significantly positive. The daily average and risk-adjusted return difference between high and low PTV portfolios is at least 0.79%. In regression analysis, I examine the relationship between the PTV constructed from the last 30 days' returns, marketwide variables and risk factors from existing literature to validate my findings. The evidence supports my hypothesis, as I discover a significant negative correlation between the PTV of the past month and expected returns.

Further, I note that although the coefficient remains negative in conditional mean models, it exhibits variation across different quantiles of conditional returns. This suggests that the pricing of PTV differs for underperforming and overperforming CCs compared to the average and during bullish or bearish phases. Moreover, I find that PTV predictions have greater robustness in smaller CCs due to their specific characteristics, such as small size, high illiquidity and high volatility. Moreover, based on my analysis, I have determined that the probability weighting (PW) component of PT has the most significant impact on predicting CC returns, suggesting a preference for lottery-like characteristics among CC investors. Further, I have found that the expected utility value does not significantly contribute to predicting CC returns. In other words, PT is a more powerful predictor of CC returns than the traditional EUT, highlighting the presence of behavioral biases such as narrow framing, loss aversion and lottery preferences among CC investors.

This study contributes to the extant literature on several accounts. First, I address a topical debate about PT- a novel strand of literature in behavioral finance- and its implications for asset pricing. This is one of the early studies on CC price dynamics from the PT perspective. The results from this study will provide a better understanding of behavioral biases among CC investors and the application of PT to the CC market. PT makes a sincere attempt to account for the irrational and systematic errors in financial decision-making, paving the way for newer investment strategies that consider investors' biases and cognitive limitations. Secondly, prior research on CC has typically utilized weekly or monthly data for analysis. However, in light of increased investments in fast trading technologies and algorithmic trading, incorporating more granular data intervals, such as daily data, improves my analysis's accuracy. Third, from a investors' perspective, I contend the usefulness of implementing a trading strategy centered around the PTV effect. Next, the study employs QR, which is more appropriate for a CC-like instrument that is evidently departed from the Gaussian distribution. Finally, the CC market's economic relevance and connectedness with the global markets are witnessing exponential growth. Moreover, as investors across countries increasingly consider CC as a potential investment avenue during stable periods and as a safe haven during the crisis, a better understanding of CC price dynamics is of great interest to policymakers, enabling the robust regulations aimed at enhancing investor confidence, ensuring price stability, protecting investors from illicit activities and money laundering.

The structure of the paper is as follows. Section 2 reviews the relevant literature on the CC market, PT, and its application in finance. Section 3 outlines the mechanics of PT, including

its components and the construction of the PTV variable. [Section 4](#) discusses the data used in the study, the empirical analysis and the findings. [Section 5](#) offers concluding remarks.

2. Literature review

2.1 *Cryptocurrency market*

The emergence of CCs has been a notable development in the digital age. The origins of CCs can be traced back to 2008 when [Nakamoto \(2008\)](#) published a white paper. The first CC, Bitcoin, was launched in 2009 and has since been followed by thousands of other CCs. Two views prevail about the CCs. The first one characterizes them as bubbles and fraudulent investments. However, the second view recognizes the potential of CCs to be an important investment asset class. The second view makes it imperative to understand the empirical asset pricing of CCs, which is the focus of this paper.

The economic relevance of the CC market is reflected in its compound annual growth rate (CAGR) of over 65% over the last decade and current market capitalization of around \$2 trillion [\[3\]](#). The CC market differs from traditional markets in several ways. First, it is decentralized, which makes it immune to various government regulations ([Cheah and Fry, 2015](#); [Urquhart, 2016](#)). Further, the 24/7 operationality of the CC market enables faster response to news and events, improving its efficiency ([Cheah and Fry, 2015](#); [Corbet et al., 2018](#)).

Moreover, the growing influence of retail investors in the financial market has undoubtedly contributed to the democratization of trading. However, this increased participation also exposes the market to greater speculation, thereby highlighting the role of investor sentiments ([Cheah and Fry, 2015](#)). Given the prevalence of retail investors and the impact of sentiments, CCs are an ideal subject for examining PT predictions, which serve as the framework for this research.

2.2 *Application of the PT in finance*

Propounded in 1979 and widely accepted as an excellent description of decision-making in laboratory settings, few well-known studies have tested PT outside in real-life situations. It's taken a long time for PT to find its application in economics and, specifically, in finance.

Though the PT finds its application across various strands of finance literature, its utility in asset pricing and explaining various market anomalies like the disposition, momentum and January effects has received substantial attention ([Barberis et al., 2021](#); [Frazzini, 2006](#)). [Benartzi and Thaler \(1995\)](#) offer the PT explanation for the equity premium puzzle. They conclude that investors are more focused on short-term losses than long-term gains, leading them to demand higher stock returns. Another study by [Benartzi and Thaler \(1999\)](#) finds that investors are more sensitive to short-term losses than long-term gains, leading them to make suboptimal investment decisions that significantly impact long-term financial outcomes.

Further, the PW component of the PT has been the focus of several studies in finance. Several studies find the skewness preferences among investors, concluding that securities' own skewness can be priced ([Barberis and Huang, 2008](#); [Boyer et al., 2010](#); [Mitton and Vorkink, 2007](#)). Using data on stock options traded on the Chicago Board Options Exchange, the USA stock market, mutual funds and commodity markets, several studies using a variety of methods to measure skewness argue that positively skewed returns or lottery-like features are appealing to investors, leading to their overvaluation ([Bali et al., 2011](#); [Boyer and Vorkink, 2014](#); [Fernandez-Perez et al., 2018](#); [Harvey and Siddique, 2000](#)).

Furthermore, studies on professional behavior have concluded that financial professionals exhibit many behavioral biases predicted by PT, including loss aversion, PW and framing effects ([Abdellaoui et al., 2013](#); [Olsen, 1997](#)). Moreover, [Teo and O'Connell](#)

(2003) argue that institutional investors display dynamic loss aversion, but the disposition effect is absent. Meaning they tend to reduce the risk following the losses and slightly increase the risk post the gains.

Benartzi and Thaler (1995) were the first to apply all PT components in financial markets. They suggest that during the editing phase, investors tend to narrow their focus and isolate information, leading them to frame a security's mental representation by examining its historical return distribution. Following their approach, Barberis *et al.* (2016) propose the same mechanics in the stock market, collecting evidence from 42 international markets. The study concludes that in cross-section, stocks with high PTV give lower subsequent returns than low PTV stocks. Similar investigations have been extended to the foreign share market (Wang *et al.*, 2021), corporate bond market (Zhong and Wang, 2018) and foreign exchange market (Kozhan *et al.*, 2020).

The PT application in understanding the CC market has received limited attention thus far. Chen *et al.* (2022) conducted a study investigating weekly data of CCs from 2014 to 2020, suggesting that CCs with high PTV exhibit lower subsequent returns in the cross-section. A similar study by Thoma (2021) also concludes that a portfolio's predictive power comes from the PTV. My study takes a new approach and differs from the earlier studies by modeling the relationship between return and PTV at different quantiles of the return distribution. This diverges from previous studies that focused solely on CCs conforming to the mean of the conditional distribution. By adopting this approach, I can uncover compelling economic relationships that might be overlooked when only considering the conditional mean. Additionally, my study assumes daily portfolio rebalancing to account for the fast-paced trading environment in CC markets. The inclusion of a large sample of CCs, representing approximately 95% of the CC market, enhances the generalizability and broader applicability of my findings.

3. PT model and construction of PTV

To measure the PTV of a CC at a time, I follow the approach suggested by Barberis *et al.* (2016) and Benartzi and Thaler (1995) and make the following assumptions. (1) The first stage is the editing or representation stage, where investors mentally represent the CC returns by looking at its historical return distribution. (2) In the second stage, i.e., the evaluation stage, investors evaluate their decisions following the PT framework. Two main components of the PT are (1) value function and (2) PW and the formula proposed by Tversky and Kahneman (1992) is:

$$PTV = \sum_{-m}^n \pi_i v(x_i), \quad (1)$$

where, the $v(\cdot)$ is the value function and π_i represent the PW function. The functional form of value function is,

$$v(x) = \begin{cases} x^\alpha, & x \geq 0 \\ -\lambda(-x)^\alpha, & x < 0 \end{cases} \quad (2)$$

The PT assigns value to the gains and losses, not the final wealth, unlike the EUT. Also, Tversky and Kahneman (1992) provides α parameter with 0.88 value that determines the S-shaped value function graph, showing how investors value gains and losses differently. The gains are to the right of the reference point, and the losses are to the left. Investors tend to be risk-averse in the domain of gains and risk-seeking in the domain of losses. The curve also represents the principle of diminishing sensitivity, meaning marginal changes will have a less

psychological impact the farther it is from the reference point. For example, in the gaining space, the increases from \$10 to \$15 have greater psychological value than an increase from \$100 to \$105.

Similarly, in the loss's domain, the loss from \$10 to \$5 will feel worse than that from \$100 to \$95. Another parameter, λ , represents the asymmetry of the two curves in the value function and captures the loss aversion behavior. If the value of λ parameter is equal to 1, meaning the sensitivity to gains and losses is the same. However, [Tversky and Kahneman \(1992\)](#) suggests λ parameter value as 2.25, which is referred to as the loss aversion ratio. The loss aversion ratio of 2.25 implies that the pain from losses is about twice the happiness from gains of the same scale. The asymmetry is visible in the abrupt change in the slope at the reference point, indicating that investors respond to losses more strongly than gains.

The second component of the PT is PW. Unlike objective probabilities used in the EUT, probabilities in the PT are decision weights. The decision weights are presented in the form of a PW function, i.e.

$$\pi_i = \begin{cases} w^+(p_i + \dots + p_n) - w^+(p_{i+1} + \dots + p_n), & 0 \leq i < n \\ w^-(p_{-m} + \dots + p_i) - w^-(p_{-m} + \dots + p_{i-1}), & -m \leq i < 0 \end{cases} \quad (3)$$

$$\text{And, } w^+(p) = \frac{p^\gamma}{(p^\gamma + (1-p)^\gamma)^{\frac{1}{\gamma}}}, w^-(p) = \frac{p^\delta}{(p^\delta + (1-p)^\delta)^{\frac{1}{\delta}}} \quad (4)$$

The functional form of the weighing function by [Tversky and Kahneman \(1992\)](#) captures the two significant effects: possibility and certainty effects. Both extreme outcomes, x_{-m} and x_n are assigned decision weight of $w^-(p_{-m})$ and $w^+(p_n)$. And if p_{-m} and p_n are small, then $w^-(p_{-m}) > p_{-m}$ and $w^+(p_n) > p_n$. Meaning investors overweigh the outcomes on both ends. Highly unlikely gain outcomes are given more weight than outcomes with moderate to high probability (possibility effect), reflecting the lottery preferences among the investors. That is, when presented with a gamble with a very small probability for large gains or a certain small amount, investors choose the gamble. On the other hand, the certainty effect means that improbable loss outcomes are given more weight than outcomes with moderate to high probability. This is a case of “will certainly happen,” and the investor will choose the certain loss outcome; this is where insurance is sold. Overweighing the tails explains investors' simultaneous demand for both lottery and insurance. Parameters δ and γ related to the PW function measure the intensity of the PW component the smaller the values for these parameters, the more the overweighing of small extreme outcomes. [Tversky and Kahneman \(1992\)](#) estimate δ and γ as 0.69 and 0.61, respectively.

The choice of reference point is a critical component of PT, and there are several options available. [Koszegi and Rabin \(2006, 2009\)](#) propose that reference points are often the expectations and beliefs about the outcome in the recent past. Excess return to a risk-free rate is one plausible option in this context. But it may not be reliable in practice because the risk-free daily rate is essentially zero nowadays. The next option is the raw return, as investors examine price charts to form expectations about the outcome in the recent past. Lastly, investors also have market performance in mind when processing investment decisions. Hence excess return over the market is also a potential option.

In the spirit of [Barberis et al. \(2016\)](#), in this study, return is the excess return to the market for my principal analysis, meaning when analyzing a CC's returns, investors do so, keeping the market return in perspective [4]. Further, while websites providing free data on CCs offer price charts for various timeframes (intraday, weekly, monthly, yearly and year-to-date), the 30-day historical window is the most commonly available and standardized option. This study assumes that investors rely on the past 30 days of daily returns when forming a mental

representation of a CC. Since the intraday data for CCs is not easily accessible to investors, daily data is the best available choice and ensures sufficient observations.

To compute the PTV at the end of the day $t-1$, I arrange the return of the past 30 ($t-30$ to $t-1$) in ascending order from most negative x_{-m} to most positive x_m . Now, these outcomes are evaluated by investors in the PT framework discussed above.

4. Empirical analysis

4.1 Data and control variables

I obtain data on CCs from coinmarketcap.com, which is widely used by CC investors and traders worldwide. It provides real-time data on CC prices, market capitalization, trading volume and other vital metrics. Drawing data from over 400 CC exchanges, the website frequently updates itself to provide timely and accurate data, making it one of the most valuable, reliable, and robust sources for CCs.

The sample period is from January 1, 2014, to December 31, 2022. Initially, there were 8,769 CCs in the sample. However, I applied the following filters to eliminate CCs that did not meet specific criteria, ensuring a robust and liquid sample set: (1) lacking data on market capitalization and trading volume data, (2) having a trading history of fewer than 180 days, (3) having an average price of less than \$0.01, (4) observations with daily returns bigger than 1000% [5]. Applying these filters results in a remaining sample size of 1,629 CCs. I utilize the daily price series to create the return, which is then used to calculate PTV daily.

I control for a set of variables considered significant in determining the cross-section of CCs, as suggested by previous research [6]. Table 1 provides the summary statistics of all the variables used in the study. It presents the time series averages of the daily cross-section. Each variable's mean and standard deviation are reported in Panel A, while Panel B reports the pairwise correlation among each pair of variables.

This study's primary variable of interest is PTV, calculated based on historical returns of the previous 30 days, as detailed in section 2. The study also addresses that the PTV variable's predictive power might be due to skewness-related variables. I address this concern by including these variables in the analysis. Some skewness-related measures considered are Max, Min, idiosyncratic skewness and co-skewness, which have been studied extensively in the cross-section of stock returns (Bali *et al.*, 2011; Boyer *et al.*, 2010; Harvey and Siddique, 2000; Walkshäusl, 2014).

Several key observations arise when examining the correlation between PTV and other variables in the context of two essential components of PT, i.e. PW and loss aversion. Firstly, PTV positively correlates with past return measures such as momentum and reversal. It also shows a positive association with various measures of skewness, including total and idiosyncratic skewness. This suggests that CCs with highly skewed returns tend to have higher PTV, indicating the influence of PW. On the other hand, PTV demonstrates a negative relationship with volatility and idiosyncratic volatility. This implies that PTV decreases for CCs with greater dispersion and volatility in the mean payoffs, reflecting the presence of loss aversion. These observations align closely with the principles of PT, providing empirical support for its postulations.

4.2 Time-series tests

The study conducts an empirical analysis of the portfolio to examine the predictive power of PTV. Portfolio analysis is valuable because it is non-parametric and not constrained by the functional relationship between PTV and future returns. Further, using portfolios enables control for noise and idiosyncratic factors, providing more robust statistical significance of

Table 1.
Average cross-sectional summary statistics

	Return	PTV	Beta	Size	Mom	Rev	Illiq	Vol	Ivol	Log Vol	SDVol	Max	Min	Kurt	Skew	Iskew	Coskew
<i>Panel A. Mean and standard deviation</i>																	
Mean	-0.001	-0.060	0.903	15.684	-0.016	-0.005	0.014	0.096	0.084	11.40	0.789	0.248	-0.221	4.823	0.192	0.418	-1.946
SD	0.104	0.047	0.803	2.562	0.370	0.217	0.049	0.067	0.069	3.503	0.481	0.214	0.196	2.701	0.898	0.935	23.289
<i>Panel B. Pearson's pairwise correlation matrix</i>																	
Return	1																
PTV	0.101	1															
Beta	-0.328	0.103	1														
Size	-0.055	0.305	0.079	1													
Mom	0.007	0.616	0.130	0.120	1												
Rev	0.359	0.246	-0.194	-0.011	-0.016	1											
Illiq	-0.026	-0.293	-0.041	-0.155	-0.053	0.005	1										
Vol	-0.034	-0.743	-0.054	-0.308	-0.162	0.025	0.397	1									
Ivol	-0.014	-0.742	-0.094	-0.325	-0.158	0.035	0.397	0.996	1								
Log Vol	-0.078	0.346	0.091	0.732	0.105	-0.027	-0.352	-0.360	-0.372	1							
SDVol	0.013	-0.230	-0.007	-0.218	0.025	0.046	0.369	0.326	0.330	-0.459	1						
Max	0.004	-0.324	0.059	-0.254	0.145	0.152	0.355	0.827	0.830	-0.267	0.281	1					
Min	0.062	0.871	0.121	0.253	0.404	0.078	-0.304	-0.882	-0.874	0.321	-0.244	-0.534	1				
Kurt	-0.001	-0.206	-0.073	-0.120	-0.109	0.031	0.050	0.395	0.397	-0.099	0.111	0.437	-0.429	1			
Skew	0.104	0.440	0.041	0.028	0.375	0.215	-0.011	-0.001	0.010	0.085	0.027	0.382	0.313	0.135	1		
Iskew	-0.008	0.445	0.076	0.166	0.340	0.165	-0.032	-0.036	-0.038	0.231	-0.022	0.297	0.293	0.115	0.827	1	
Coskew	0.296	0.073	0.203	-0.035	0.065	-0.01	-0.023	-0.078	-0.067	-0.034	0.031	0.008	0.109	-0.035	0.081	-0.126	1
Note(s): Panels A & B present daily cross-sectional summary statistics of time-series averages. Panel A shows the mean and standard deviations, while Panel B reports Pearson's correlation matrix. PTV is calculated based on historical returns of the previous 30 days, as detailed in section 2 . Beta is estimated as the slope of regressing a CC's daily log excess return over the risk-free rate on its market excess return in the last 30 days. Illiquidity is Amihud's (2002) measure of illiquidity, scaled by 10 ⁵ . Volume is the natural logarithm of CC's mean daily trading volume in the previous 30 days. SD Vol is the standard deviation of a CC's daily trading volume in the last 30 days. Vol (volatility) is the standard deviation of CC's daily return in t-30 days. Ivol is the idiosyncratic volatility of CC daily return (Ang et al., 2006). MAX represents a CC's maximum one-day return in t-30 days (Bali et al., 2011), while MIN represents a CC's minimum one-day return in t-30 days (Bali et al., 2011). Mom (momentum) is a CC's cumulative return from day t-30 to t-8. Rev (reversal) is a CC's cumulative return from day t-8 to t-1. Kurtosis is the kurtosis of a stock's daily return over the previous 30 days. Skewness is the skewness of a stock's daily return over the last 30 days. We compute idiosyncratic and co-skewness (systematic skewness) in the manner proposed by Harvey and Siddique (2000)																	

systematic factors. Further, by creating portfolios based on specific characteristics, I can decrease portfolio variance and observe differences in average returns more sharply.

I undertake the portfolio sort approach as follows. Each day, I sort the CCs into decile portfolios based on their PTV. Decile 1 (low PTV) is the portfolio of CCs with the lowest PTV on the previous day, and decile 10 (high PTV) is the portfolio of CCs with the highest PTV on the previous day. I assume that investors hold the portfolio for only one day and calculate the one-day ahead equal-weighted and value-weighted mean returns for each PTV-sorted decile. Using this time series of daily returns, I calculate each PTV-sorted portfolio's average return, CAPM alpha and three-factor alpha. Table 2 reports the average return in excess of the risk-free rate, CAPM alpha from the regression on excess market return, and three-factor alpha from the regression on market factor, size factor (SMB), and momentum factor (WML) for each decile portfolio. To test my hypothesis, the last column reports the difference between decile 1 (low) and decile 10 (high). The results reported in Table 2 show that there is a significant positive difference between decile 1 (low) and decile 10 (high) across all scenarios.

Further, the decline of average return, CAPM, and three-factor alpha from decile 1 to 10 is much more consistent in equal-weighted portfolios than in value-weighted. In value-weighted portfolios, decile 10 witnesses a sudden upsurge and greater average return and CAPM alpha than decile 9. However, in summary, my findings indicate a significant negative relationship between PTV and returns, with lower PTV portfolios generating higher returns and alpha compared to higher PTV portfolios in both equal-weighted and value-weighted portfolios.

Table 2 and Figure 1 show that the three-factor alphas decline almost linearly from the low PTV portfolio to the high PTV portfolio, except for a significant drop in decile two. Figures A1 and A2 show that the average excess return and the CAPM alpha follow a similar pattern from decile 1 to 10. These observations suggest CCs with high mean payoffs and low volatility attract investors, leading to their overvaluation and lower returns. Therefore, my hypothesis that the PT model has significant predictive power in understanding the dynamics of CCs is supported.

Furthermore, while the difference in average return between low PTV and high PTV portfolios is economically and statistically significant for both value-weighted and equal-weighted returns, it is more significant for equal-weighted returns. This supports my other claim that PTV is more significant in small CCs than large ones.

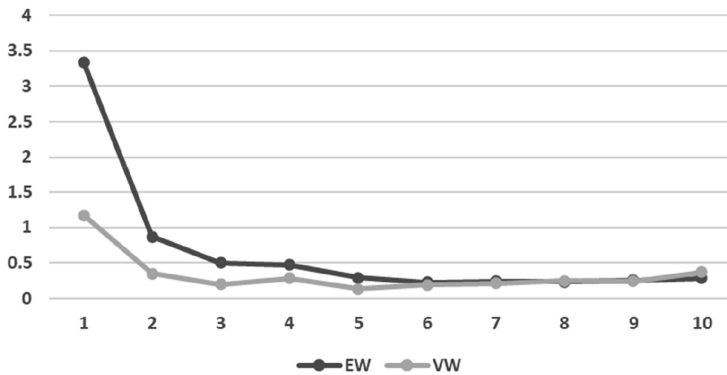
In addition, I investigate whether PTV has predictive power on CC returns beyond the day of portfolio construction. To test this, I now sort CCs in decile portfolios based on the PTV of $t-1$ and look at returns not only on day t but also on day $t+1$, $t+2$ to $t+6$. Figure 2 corresponds to equal-weighted and value-weighted three-factor alphas, illustrating that PTV can predict CC returns for several days after portfolio construction. The long-short strategy alpha falls linearly from t to $t+6$, except for equal-weighted portfolios, where the three-factor alpha declines sharply on $t+2$. This pattern can be attributed to short-term reversal. However, in further analysis, I confirm that PTV retains its predictive power even after I control for reversal in the model. Figures A3 and A4 depict similar findings for average excess return and CAPM alpha. Overall, my results suggest that the PT model has significant predictive power in understanding the cross-sectional dynamics of CCs.

4.3 Robustness of time-series findings

Table 3 presents robustness checks for the portfolio sort results. The two right-most columns display the three-factor alphas for the low-PTV minus high-PTV portfolio based on equal or value-weighted returns. In my primary analysis, I construct the PTV variable using the past 30 days returns. However, I find consistent results when using different time windows for PTV construction (90, 60, 15 and 7 days). Additionally, when constructing the past return

Table 2.
Univariate portfolio
analysis

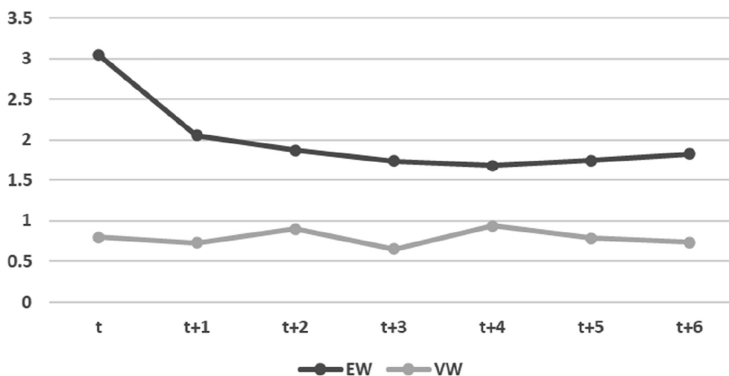
	Low	PTV2	PTV3	PTV4	PTV5	PTV6	PTV7	PTV8	PTV9	High	L-H
<i>Excess return</i>											
EW	3.424*** (20.826)	0.910*** (7.106)	0.503*** (4.217)	0.505*** (3.629)	0.296*** (2.776)	0.229*** (1.995)	0.227* (1.931)	0.216* (1.923)	0.246*** (2.260)	0.296** (2.541)	3.125*** (18.753)
VW	1.137*** (5.544)	0.320** (2.247)	0.136 (1.102)	0.226* (1.897)	0.077 (0.713)	0.113 (0.970)	0.147 (1.437)	0.176* (1.758)	0.200** (2.052)	0.335*** (3.186)	0.799*** (3.642)
<i>CAPM alpha</i>											
EW	3.472*** (21.785)	0.921*** (7.388)	0.517*** (4.832)	0.521*** (4.039)	0.303*** (2.975)	0.233*** (2.210)	0.240** (2.318)	0.227** (2.128)	0.258*** (2.915)	0.305*** (2.881)	3.168*** (19.636)
VW	1.180*** (5.698)	0.335** (2.420)	0.155 (1.272)	0.245** (2.063)	0.090 (0.824)	0.126 (1.095)	0.170 (1.626)	0.192* (1.933)	0.212** (2.322)	0.352*** (3.390)	0.828*** (3.901)
<i>Three factor alpha</i>											
EW	3.346*** (22.864)	0.874*** (7.791)	0.501*** (4.734)	0.470*** (4.734)	0.292*** (2.978)	0.230** (2.242)	0.243** (2.390)	0.237** (2.251)	0.256*** (2.907)	0.286*** (2.785)	3.060*** (21.121)
VW	1.173*** (5.769)	0.353*** (2.590)	0.199* (1.670)	0.284** (2.462)	0.136 (1.235)	0.192* (1.649)	0.217** (2.101)	0.248** (2.458)	0.245*** (2.658)	0.372*** (3.283)	0.802*** (3.825)
Note(s): The table reports the average daily excess return and daily alphas (CAPM and three-factor alpha) on equal-weighted and value-weighted PTV sorted deciles. In the last column, the table shows the average excess return, CAPM alpha and three-factor alpha for a zero-cost long-short portfolio that takes a long position in decile 1 (low PTV) and a short position in decile 10 (high PTV). HAC-robust <i>t</i> -statistics based on Newey–West standard errors, are displayed in parentheses. Statistical significance is denoted by *, **, and ***, representing significance levels at 10, 5 and 1 %, respectively											



Note(s): The figure illustrates the three-factor alpha for each portfolio sorted by PTV. We rank the CCs into deciles based on their daily PTV to create these portfolios. Decile 1 represents CCs with the lowest PTV from the previous day, while decile 10 represents CCs with the highest PTV from the previous day. Our analysis assumes that investors hold the portfolio for only one day. Consequently, we calculate the one-day ahead equal-weighted and value-weighted mean returns for each PTV-sorted decile. We calculate the three-factor alpha for each PTV-sorted portfolio using this time series of daily returns

Source(s): Author's own work

Figure 1.
Performance of PTV
deciles



Note(s): The figure showcases the three-factor alpha, calculated on both an equal-weighted (EW) and value-weighted (VW) basis, of a long-short portfolio that buys (shorts) CCs that were in the lowest (highest) PTV decile in the past. The vertical axis represents the daily alpha, which measures the portfolio's risk-adjusted performance, while the horizontal axis represents the time lag, in days, between constructing the portfolio and calculating the returns

Source(s): Author's own work

Figure 2.
Long-short portfolio
returns over time

distribution for CCs, I consider returns in excess of the market return. However, the results in Table 3 demonstrate similar outcomes when using raw returns and returns in excess of the risk-free rate instead. These findings reinforce the robustness of my results, highlighting the suitability of PT as an appropriate approach to model CC investor behavior.

Table 3.
Robustness

		EW	VW
Window for constructing PTV	Past 90 days	2.458*** (16.847)	0.724*** (3.974)
	Past 60 days	2.122*** (14.527)	0.529*** (2.958)
	Past 15 days	2.996*** (17.499)	0.583*** (2.672)
	Past 7 days	3.724*** (20.155)	0.650*** (3.252)
	Raw return	3.002*** (19.266)	0.754*** (3.805)
Other return measures (Reference point)	Returns in excess of the risk-free rate	3.002*** (19.277)	0.753*** (3.801)

Note(s): We test the robustness of the results by varying the length of the time window used to construct the PTV variable. For the benchmark analysis, the time window was 30 days. We further test the robustness on 90, 60, 15 and 7 days time windows. Additionally, the reference point in our benchmark analysis was assumed to be excess return over the market. We test our findings on raw return and return over the risk-free rate as the reference point. Statistical significance is denoted by *, **, and ***, representing significance levels at 10%, 5%, and 1%, respectively

4.4 Regression analysis

Despite several advantages, portfolio analysis has its limitations. One potential concern is that a CCs' return may be correlated with other characteristics, meaning that portfolio analysis results may be influenced by factors other than the PTV. To address this, I employ regression methodology and use heteroscedasticity and autocorrelation consistent (HAC) robust standard errors for computing *t*-statistics. This approach offers a more nuanced understanding of the factors driving investment performance beyond the PTV. Table B4 provides OLS-based, panel-fixed effect and QR estimates [7]. The estimates based on the conditional mean estimation reinforce the findings from portfolio analysis, indicating an inverse relationship between PTV and CC returns. This aligns with previous studies by Barberis *et al.* (2016), Chen *et al.* (2022), and Wang *et al.* (2021) that have observed a similar mean relationship between returns and PTV. However, it is essential to acknowledge that CC data exhibit non-linearities, skewness and excess kurtosis, which limit the insights derived solely from estimations based on the conditional mean. To overcome this limitation, I employ a QR approach, following the methodology proposed by Koenker and Bassett (1978) and Koenker and Hallock (2001). This approach allows for a more comprehensive and robust analysis, considering the non-linear characteristics of CC data and providing a deeper understanding of the relationship between returns and PTV across various quantiles.

QR has gained increasing usage in recent empirical finance literature. In asset pricing literature, Aziz and Ansari (2018), Barnes and Hughes (2002), and Nath and Brooks (2015) applied QR to test the predictions of risk factors over the distribution of returns. Using QR, I model the relationship between return and PTV, across quantiles, for CCs that overperform and underperform relative to the mean [8]. Table 4 reports regression estimates with several control variables. I estimate the following model for different values of τ ($\tau = 0.05, 0.10, 0.20, \dots, 0.95$):

$$y = X\beta(\tau) + \epsilon \tag{5}$$

here, $y_{[n]}$ is the vector of responses, $X_{n \times p}$ is the regressor matrix, $\beta_{[p]}(\tau)$ is the vector of unknown parameters for the generic conditional quantile τ and $\epsilon_{[n]}$ is the vector of unknown errors. The QR coefficient at a given quantile indicates the effect of a unit change in an explanatory variable on the response variable when other explanatory variables are held

	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0.95
PTV	1.3038*** (124.430)	0.922*** (119.150)	0.538*** (114.610)	0.3134*** (93.720)	0.1546*** (59.290)	0.0367*** (20.620)	-0.0502*** (-23.78)	-0.1936*** (-61.68)	-0.4423*** (-91.04)	-0.9312*** (-113.71)	-1.4988*** (-117.94)
Beta	-1.1786*** (-31.31)	-1.1214*** (-45.76)	-0.9008*** (-59)	-0.671*** (-61.89)	-0.4333*** (-50.26)	-0.232*** (-18.25)	0.222*** (31.740)	0.5361*** (51.05)	0.817*** (55.230)	1.0725*** (38.600)	1.1373*** (24.760)
Size	0.2128*** (15.740)	0.1778*** (20.340)	0.1349*** (25.280)	0.1012*** (26.060)	0.0679*** (23.080)	0.0292*** (13.430)	-0.0146*** (-5.95)	-0.0617*** (-17.59)	-0.1161*** (-23.02)	-0.2039*** (-23.19)	-0.3284*** (-22.45)
Momentum	-0.0604*** (-85.74)	-0.0437*** (-89.28)	-0.0262*** (-78.27)	-0.0157*** (-58.43)	-0.0082*** (-35.13)	-0.0018*** (-9)	0.0039*** (17.520)	0.0122*** (44.00)	0.0253*** (68.660)	0.0492*** (86.850)	0.0764*** (86.160)
Reversal	-0.0952*** (-93.39)	-0.0753*** (-98.66)	-0.0544*** (-94.14)	-0.0408*** (-83.96)	-0.0302*** (-67.64)	-0.0217*** (-51.02)	-0.0182*** (-43.93)	-0.0131*** (-27.41)	-0.0024*** (-3.99)	0.0209*** (26.320)	0.049*** (41.470)
Illiquidity	-0.2522*** (-19.14)	-0.0879*** (-7.55)	0.0058 (1.270)	0.0196*** (13.180)	0.0164*** (16.490)	0.007*** (11.420)	-0.0007 (-0.76)	-0.0052*** (-2.92)	0.008*** (1.980)	0.0804*** (9.090)	0.187*** (31.600)
Log volume	-0.0029 (-0.27)	-0.0358*** (-51.8)	-0.0421*** (-9.9)	-0.0365*** (-12.08)	-0.0283*** (-12.56)	-0.0185*** (-11.44)	-0.0066*** (-3.41)	0.0055*** (1.980)	0.0276*** (6.830)	0.062*** (8.920)	0.1162*** (9.820)
SD volume	-0.8529*** (-12.51)	-0.5347*** (-14.72)	-0.2487*** (-11.79)	-0.0975*** (-7.04)	-0.0007 (-0.07)	0.01 (1.400)	-0.011 (-1.21)	0.0666*** (5.220)	0.2253*** (11.030)	0.5998*** (15.810)	1.0967*** (14.700)
Kurtosis	-0.1385*** (-15.77)	-0.0016 (-0.33)	0.0455*** (16.700)	0.0429*** (22.410)	0.0216*** (14.590)	-0.0027*** (-2.48)	-0.0323*** (-7.63)	-0.0641*** (-4.03)	-0.0851*** (-11.06)	-0.05*** (3.600)	0.116*** (2.880)
Skewness	-0.3916*** (-13.04)	-0.2591*** (-14.25)	-0.153*** (-13.7)	-0.1073*** (-13.2)	-0.0679*** (-10.1)	-0.0461*** (-8.19)	-0.0447*** (-7.63)	-0.0295*** (-4.03)	0.0049 (0.490)	0.0612*** (3.600)	0.0921*** (2.880)
Idio-skew	-1.2896*** (-44.16)	-0.934*** (-51.78)	-0.5749*** (-50.72)	-0.3547*** (-43.77)	-0.1949*** (-28.94)	-0.0516*** (-9.08)	0.0695*** (11.720)	0.2114*** (28.540)	0.4033*** (39.340)	0.7762*** (45.830)	1.202*** (39.950)
Co-skew	-0.0117*** (-8.87)	-0.0062*** (-7)	-0.0023*** (-3.68)	-0.0005 (-1.31)	0.0003 (0.630)	0.0004* (1.680)	0.0002 (0.800)	0.0011** (2.480)	0.0032*** (6.390)	0.0078*** (6.600)	0.014*** (8.910)
Goodness of fit	9.30%	6.10%	3.30%	1.70%	0.80%	0.30%	0.50%	1.30%	2.80%	6.20%	10.20%

Note(s): This table presents the results of quantile regressions with various control variables. The dependent variable in all specifications is the next day's return of the respective CC. Columns 2 to 12 are estimates across different quantiles (0.05–0.95). HAC-robust *t*-statistics are in parenthesis. Beta, size, kurtosis, skewness, idiosyncratic and co-skewness measures are scaled by 10^2 . We use the “Quantreg” package in “R” to implement the model.

$r_{i,t} [Q_{\tau}(\tau) PTV_{i,t-1}, Size_{i,t-1}, Momentum_{i,t-1}, Reversal_{i,t-1}, Illiq_{i,t-1}, Logvol_{i,t-1}, Skew_{i,t-1}, Kurt_{i,t-1}, SDvol_{i,t-1}, Coskew_{i,t-1}]$

$= \beta_0(\tau) + \beta_1(\tau) PTV_{i,t-1} + \beta_2(\tau) Size_{i,t-1} + \beta_3(\tau) Momentum_{i,t-1} + \beta_4(\tau) Reversal_{i,t-1} + \beta_5(\tau) Illiq_{i,t-1} + \beta_6(\tau) Logvol_{i,t-1} + \beta_7(\tau) Skew_{i,t-1} + \beta_8(\tau) Kurt_{i,t-1} + \beta_9(\tau) SDvol_{i,t-1} + \beta_{10}(\tau) Coskew_{i,t-1} + \beta_{11}(\tau) Skew_{i,t-1} + \beta_{12}(\tau) Coskew_{i,t-1}$

and 1%, respectively

Table 4.
Quantile regression
analysis

constant. Earlier studies show that returns are inversely related to PTV, so I expect a significantly negative coefficient on PTV across all quantiles. However, I find that the coefficient on PTV varies at different points along the distribution of conditional returns, indicating that PTV is priced differently for CCs that underperform or overperform relative to the mean. Therefore, I cannot assume that PTV explains the entire cross-section of returns in a linear fashion.

When examining the relationship between PTV and CCs performance, in [Figure A5](#), I observe an interesting dynamic pattern that resembles an inverse “S-shaped” curve. Specifically, I find a positive coefficient for CCs performing below the mean and a negative coefficient for CCs outperforming. Notably, the negative coefficient is significant for the top 50 percentile of CC returns, indicating that the overvaluation hypothesis suggested by earlier studies holds only for CCs that outperform the average. One possible explanation is that CCs in the upper quantiles possess attractive characteristics, such as lottery-like payouts, high abnormal trading volume, low volatility and dispersion, which attract significant positive attention [\[9\]](#). Conversely, CCs in the lower quantiles exhibit unobserved characteristics that deviate from PTV predictions and instead demonstrate momentum. This implies that investors tend to underreact to CCs in these quantiles, leading to performance persistence [\[10\]](#). Consistent with [Zuchel and Weber \(2001\)](#), I argue that under-reaction in these quantiles results from the disposition effect evident in the investors. Furthermore, as I approach the extremes (tails) of the distribution, the influence of CC characteristics becomes increasingly evident, as indicated by the highest PTV coefficient. This suggests that investor behavior is particularly pronounced during instances of extreme losses or gains.

Furthermore, given the extensive research conducted on skewness-related measures in the field of cross-sectional returns, I incorporate them as control variables in my model. Notably, even after their inclusion, PTV retains its significance, highlighting the robustness of PTV as a factor. The findings reinforce the notion that the PTV is an important factor and has significant predictive power. Moreover, it can effectively analyze and comprehend the dynamics of cross-sections in CCs.

In the subsequent sections of the paper, I employ the panel fixed effect model to explore the dependence and to gain insights into the relationship. Following the approach of [Chen *et al.* \(2022\)](#), my interpretation is as follows: over time, as the PTV of a CC increases (decreases) relative to the average PTV of active CCs in the cross-section, its subsequent return falls (rises) relative to the average returns of all CCs in the cross-section.

4.5 Characteristics of PTV deciles

In the previous sections, I established that PTV has significant predictive power along the cross-section of CC returns. In line with [Barberis *et al.* \(2016\)](#), I assumed that investors make the mental representation of a CC's past return distributions and then evaluate it in the PT framework. The results reveal an average inverse relationship between PTV and CC returns. CCs that investors find appealing from the PT perspective form part of the portfolio. The increased demand for such CCs leads to overvaluation and lower subsequent returns. Conversely, CCs that are unappealing under the PT framework become undervalued and earn higher subsequent returns, reflecting compensation for the low mean payoffs and high volatility of such CCs.

In this section, I examine the characteristics of low and high PTV CCs and the return distribution of high PTV CCs. What factors do investors get attracted to? To answer this question, each day, I sort CCs into deciles based on PTV, and then for each decile, I look at the average cross-section value of each characteristic. I then compute the mean of the daily average decile characteristic. Based on PT postulations, I expect the high PTV deciles to exhibit the following features: high past returns, low volatility and high skewness.

The results presented in Table 5 largely support the general PT components. High PTV deciles have higher average past returns than low PTV deciles (columns 12 and 13). Their volatility is significantly lower (columns 8 and 9), confirming the loss aversion component of PT, which suggests a preference for lower volatility investments. Additionally, the skewness of the high PTV deciles is significantly higher than low PTV deciles (columns 15 and 16), indicating that investors in high PTV deciles tend to overweight the probability of extreme positive outcomes, consistent with lottery preferences. My analysis complements the observations from the correlation matrix, where PTV positively relates to past return measures like momentum and reversal and various skewness measures while negatively associated with volatility and idiosyncratic volatility.

4.6 *The role of limits to arbitrage*

In this section, I test the prediction that PTV has more predictive power in smaller CCs that are difficult to arbitrage. To do so, I use several proxies for limits to arbitrage, such as market capitalization, volatility, idiosyncratic volatility and illiquidity, as suggested in previous studies. The results from regressions and portfolio analysis are provided in Table 6. In panel A, proxies of limits to arbitrage interact with PTV, and new independent variables are added to the model. My findings indicate that the coefficient on the interaction term $PTV \times Size$ is significant at the 1% level, while $PTV \times Volatility$ and $PTV \times IVOL$ are significant at the 5% level, suggesting that PTV has a stronger predictive ability for small and volatile CCs. The signs on these coefficients support my prediction that investors' irrationality drives the relationship between PTV and subsequent returns. Further, as expected, the sign on $PTV \times Illiquidity$ is negative, meaning PTV is stronger among illiquid CCs but is not statistically significant.

In Panel B, I employ a two-step approach. Firstly, I divide CCs into five quantiles based on each proxy variable. I further sort the CCs into five quantiles within each of these quantiles based on their PTV. Subsequently, I calculate the differences in alpha values between the first and last quantiles within each subquantile. The results here support my analysis as I make the following observations. For example, in size proxy, the difference between low and high PTV is the largest and most significant in portfolio one, representing the small CCs. I drew similar conclusions from other proxy analyses, supporting my claim that PTV is more robust in CCs that are less affected by arbitrage forces.

4.7 *Components of PT*

This section focuses on investigating which component of PT (loss aversion, PW or curvature) is the most significant in explaining CC price dynamics. My analysis of the decile portfolios in Table 2 suggests that skewness may play a crucial role, as it increases monotonically from decile 1 to 10. To test this hypothesis, I conducted a regression analysis with seven different models, each focusing on a specific component of PTV, as shown in Table 7. Each model includes one or more PTV components and other well-known predictors. For instance, in Model 1, the loss aversion component of PTV is activated, with an assigned parameter value of $\lambda = -2.25$, while all other components (α, γ, δ) are set to one instead of their original values of 0.88, 0.61 and 0.69. Models 2 and 3 are defined similarly. In Model 4, only the loss aversion and curvature components are active, and the PW component (γ, δ) is set to one instead of 0.61 and 0.69, respectively. This process is repeated for columns 5 and 6. Through these models, I aim to determine which PTV component has the most substantial effect on future returns.

In line with Barberis *et al.* (2016), I analyzed the *t*-statistics of each PTV component in my models to determine which component has the most significant impact on future returns. The results indicate that the PW component is the dominant factor driving the predictive power of PTV. Models 2, 5, and 6, which include the PW component, exhibit the highest *t*-statistics.

Table 5.
Characteristics of
portfolio deciles

Port	PTV	Beta	Size	Illiq	Log Vol	SD Vol	Volatility	Ivol	Max	Min	Mom	Reversal	Kurtosis	Skew	Iskew	Coskew
1	-0.157	0.858	13.463	0.058	8.390	1.084	0.204	0.196	0.487	-0.567	-0.289	-0.103	5.482	-0.249	-0.219	0.387
2	-0.093	0.855	14.35	0.020	9.727	0.880	0.116	0.106	0.282	-0.286	-0.163	-0.056	4.613	-0.019	0.064	-3.331
3	-0.074	0.913	14.881	0.013	10.426	0.812	0.097	0.085	0.236	-0.231	-0.102	-0.041	4.469	0.032	0.162	-2.446
4	-0.063	0.892	15.296	0.010	10.976	0.772	0.086	0.074	0.213	-0.200	-0.062	-0.028	4.454	0.084	0.258	-1.518
5	-0.054	0.908	15.600	0.008	11.411	0.739	0.079	0.067	0.198	-0.179	-0.029	-0.017	4.439	0.124	0.353	-2.599
6	-0.047	0.909	15.902	0.007	11.810	0.719	0.075	0.062	0.192	-0.165	0.006	-0.005	4.496	0.167	0.445	-2.334
7	-0.04	0.910	16.265	0.006	12.234	0.702	0.070	0.058	0.185	-0.152	0.039	0.007	4.550	0.210	0.524	-2.927
8	-0.033	0.907	16.632	0.005	12.650	0.692	0.067	0.054	0.180	-0.140	0.071	0.022	4.603	0.260	0.607	-1.443
9	-0.025	0.894	17.057	0.005	13.148	0.702	0.067	0.054	0.191	-0.130	0.130	0.049	4.868	0.412	0.775	-1.778
10	-0.006	0.942	17.671	0.007	13.913	0.789	0.084	0.071	0.287	-0.131	0.328	0.151	6.191	0.879	1.174	-1.136

Note(s): To analyze the characteristics of CCs, we employ a daily sorting approach based on PTV. Each day, we sort the CCs into deciles based on their PTV. We calculate the average value of the variable listed in the top row for each decile, considering all the CCs within that decile. Afterward, we take the averages of these mean characteristic values across all days in the sample period. The specific variables used in this analysis are defined in [Table 2](#)

Panel A				
PTV	−0.5368*** (−6.311)	−0.2084*** (−22.878)	−0.212*** (−22.164)	−0.2125*** (−22.125)
PTV*Size	1.9866*** (3.692)			
PTV*Vol		−0.0148 (−1.585)		
PTV*IVOL			−0.0211** (−2.184)	
PTV*Illiq				−0.0222** (−2.305)
Beta	−0.1163*** (−3.492)	−0.0834*** (−3)	−0.0865*** (−3.112)	−0.0862*** (−3.099)
Size	−0.2058*** (−6.787)	−0.3282*** (−21.552)	−0.3286*** (−21.729)	−0.3287*** (−21.726)
Mom	0.012*** (17.303)	0.011*** (16.869)	0.0113*** (18.180)	0.0113*** (18.203)
Reversal	−0.0272*** (−9.641)	−0.0284*** (−9.678)	−0.0281*** (−9.671)	−0.0281*** (−9.66)
Illiquidity	−0.0024 (−0.62)	−0.0063* (−1.846)	−0.0018 (−0.706)	−0.0019 (−0.723)
Volatility	0.0015 (0.097)	−0.0161 (−1.143)	−0.0134 (−0.932)	−0.0118 (−0.815)
Ivol	−0.1836*** (−11.551)	−0.1181*** (−9.359)	−0.1314*** (−9.557)	−0.1337*** (−9.465)
Max	2.5381*** (6.714)	1.5639*** (6.028)	1.7853*** (6.153)	1.8007*** (6.186)
Min	1.0358*** (3.118)	0.6732*** (2.675)	0.8273*** (2.852)	0.8382*** (2.885)
Panel B				
	Size	Volatility	IVOL	Illiquidity
Portfolio 1	5.1883*** (24.147)	−0.1800** (−2.449)	−0.1218 (−1.597)	−0.2350*** (−3.417)
Portfolio 2	0.9783*** (8.147)	0.1462** (2.125)	0.1462** (2.125)	0.1119* (1.781)
Portfolio 3	0.2024* (1.913)	0.2541*** (3.103)	0.2493*** (2.905)	0.2363*** (2.901)
Portfolio 4	0.0329 (0.409)	0.9647*** (9.958)	0.7974*** (8.233)	0.6172*** (5.829)
Portfolio 5	−0.2973*** (−3.644)	5.6947*** (24.936)	5.6992*** (25.207)	5.5755*** (22.972)

Note(s): Panel A reports the results of regressions of CC returns on PTV–1 the prospect theory value of a CC's historical return distribution from day $t-30$ to $t-1$ – and its interaction with four variables that capture limits to arbitrage: Size, the log market capitalization (scaled by 10^2); Amihud's (2002) measure of illiquidity; volatility and Ivol, idiosyncratic return volatility, as in Ang *et al.* (2006). The dependent variable in all specifications is the next day's CC return. In panel B, first, we create five quantiles based on each proxy variable. Then, we further sort into five quantiles based on the PTV within these quantiles. The alpha differences between each subquantile's first and last quantile are reported here. Statistical significance is denoted by *, **, and ***, representing significance levels at 10%, 5%, and 1%, respectively

Table 6.
Limits to arbitrage
and PTV

Moreover, adding the PW component in Models 4 and 7 significantly improves the t -statistics. This finding confirms my earlier observation that the PW component plays a crucial role in the predictive power of PTV.

Table 7.
Different components
of PTV

Active component	1 LA	2 PW	3 CC	4 LA/CC	5 LA/PW	6 CC/PW	7 LA/CC/PW
PTV	-0.2971*** (-18.915)	-0.4906*** (-19.493)	0.3184*** (-18.518)	-0.1876*** (-18.804)	-0.2513*** (-20.615)	-0.3684*** (-22.9)	-0.2064*** (-23.078)
Beta	-0.075*** (-2.687)	-0.1246*** (-4.377)	-0.0855*** (-3.115)	-0.0728*** (-2.685)	-0.0905*** (-3.164)	-0.1052*** (-3.769)	-0.0851*** (-3.066)
Size	-0.3362*** (-22.017)	-0.3353*** (-22.159)	-0.3418*** (-22.472)	-0.3384*** (-22.006)	-0.3274*** (-22.223)	-0.3313*** (-22.18)	-0.3264*** (-21.612)
Mom	0.0144*** (21.947)	0.012*** (18.692)	0.0122*** (18.837)	0.0119*** (18.521)	0.0115*** (18.825)	0.0114*** (18.667)	0.011*** (17.147)
Reversal	-0.0251*** (-9.099)	-0.0278*** (-10.222)	-0.0274*** (-9.554)	-0.0275*** (-9.307)	-0.0282*** (-9.963)	-0.0282*** (-9.74)	-0.0283*** (-9.653)
Illiquidity	-0.0803 (-0.318)	-0.2958 (-1.032)	-0.2194 (-0.8)	-0.0074 (-0.029)	-0.3651 (-1.432)	-0.2935 (-1.15)	-0.1159 (-0.476)
Volatility	-0.0604*** (-3.979)	0.0852*** (5.612)	0.0331** (2.201)	-0.0586*** (-4.038)	0.0269* (1.829)	0.0459*** (3.116)	-0.0171 (-1.211)
IVOL	-0.1098*** (-8.465)	-0.0859*** (-6.738)	-0.0107 (-0.841)	-0.055*** (-4.392)	-0.164*** (-12.187)	-0.04*** (-3.219)	-0.1149*** (-9.168)
MAX	0.7017*** (2.729)	3.5262*** (10.389)	-1.1195*** (-4.781)	-0.1977 (-0.866)	2.2467*** (7.944)	1.8077*** (6.705)	1.4551*** (5.861)
MIN	-1.3873*** (-5.734)	3.1137*** (9.022)	0.0928 (0.332)	-1.4738*** (-5.999)	1.9755*** (7.228)	1.7401*** (6.154)	0.5898** (2.395)

Note(s): The table reports regressions results for different PTV components. The dependent variable in all specifications is the one-day ahead return. The variation across the seven models lies in the active PTV component. LA here represents the loss aversion component, PW is the probability weighting component and CC is the concavity/convexity component. In models 1 to 3, only one component of PTV is active at a time. For example, in model 1, the “LA” designation indicates that the loss aversion component is active and incorporated into the PTV variable. In the subsequent three models, two components are simultaneously active. In the final model, the PTV variable encompasses all its components, representing the entirety of the PTV used in the study’s analysis (as presented in Table 4). Statistical significance is denoted by *, **, and ***, representing significance levels at 10%, 5%, and 1%, respectively

To further investigate this component, in Table B5, I ran 11 different models, with only the PTV variable changing across all specifications, while all other control variables remained the same. Each model had a different parameter value for the PW component, ranging from 0.31 and 0.39 to 1.31 and 1.39. However, the loss aversion and concavity/convexity components had the same parameter values as before ($\alpha = 0.88$ and $\lambda = -2.25$). The results suggest that PTV is most significant for PW parameters being less than 1 which suggests overweighing of tails, compared to PW parameter being greater than 1, which refers to underweighing of tails. Again, this supports my earlier claim that PW plays a significant role in my results. The findings align with studies claiming that securities' own skewness is a priced factor (Barberis and Huang, 2008; Boyer *et al.*, 2010; Harvey and Siddique, 2000).

4.8 PTV vs EUT

Although there have been significant advancements in behavior and decision-making science, the EUT remains one of the most commonly used paradigms for understanding decision-making under risk and uncertainty. This study assumes that investors evaluate outcomes using the PT framework after observing historical return distribution. I test if the rational decision-making model, EUT, significantly predicts CC returns. Similar to Barberis *et al.* (2016) and Zhong and Wang (2018), I assume that investors attempt to maximize their expected utility and use a constant relative risk aversion (CRRA) utility function, similar to:

$$EU = \frac{1}{30} \sum_{i=-m}^n \frac{(1 + r_i)^{1-\theta}}{1 - \theta} \quad (6)$$

where $r_{-m}, \dots, r_n$ are CC's daily returns over the last 30 days. There is a wide range of coefficients of relative risk aversion in the literature (Gandelman and Hernandez-Murillo, 2013). I consider various values for θ , the risk aversion parameter, ranging from 0.5 to 10. Table 8 shows no evidence supporting EUT as a significant predictor. In panel A, EUT is only significant in deciles 2 and 3. Panel B confirms these results, where EUT is insignificant in most cases, except for $\theta = 0.5$ & $\theta = 1$, whereas, the coefficient on PTV is consistently negative and statistically different from zero at the 1% level. The results validate my claim that investors in the CC market demonstrate bounded rationality, including traits such as loss aversion and a preference for lottery-like investments.

5. Conclusion

Conventionally, academic literature has predominantly utilized EUT and rational equilibrium models with adjustments for behavioral biases to model individual investment decision-making. However, recent research indicates that investors often exhibit irrational and unsophisticated decision-making, particularly in the CC market, where prices are primarily driven by investor sentiments due to the lack of fundamental value. As a result, behavioral models that consider bounded rationality and investor biases are gaining popularity. Among these models, PT has received significant attention.

By analyzing a large sample of 1,629 CCs, this study demonstrates an average inverse relationship between PTV and CC returns. However, this relationship exhibits asymmetry across different quantiles, indicating that investor reactions vary based on market conditions. Specifically, a negative relationship is observed between PTV and CC future returns in higher quantiles. Conversely, CCs in lower quantiles exhibit momentum in PTV instead. Moreover, PTV provides more robust predictions for smaller CCs characterized by high volatility and illiquidity. Notably, the findings highlight the dominant role of the PW component in PT for predicting CC behaviors, suggesting a preference for lottery-like characteristics among CC investors.

Table 8.
PTV vs EU

	$\theta = 0.5$	$\theta = 2$	$\theta = 3$	$\theta = 4$	$\theta = 5$	$\theta = 6$	$\theta = 7$	$\theta = 8$	$\theta = 9$	$\theta = 10$
<i>Panel A. Portfolio analysis</i>										
Low	0.341*** (2.913)	1.053*** (5.386)	0.912*** (5.217)	0.752*** (4.678)	0.77*** (4.507)	0.586*** (3.492)	0.522*** (3.224)	0.602*** (3.452)	0.57*** (3.401)	0.526*** (3.133)
PTV2	0.285*** (2.612)	0.382*** (3.048)	0.295*** (2.233)	0.457*** (3.218)	0.270*** (1.979)	0.258* (1.909)	0.224 (1.629)	0.256* (1.889)	0.234* (1.703)	0.347*** (2.386)
PTV3	0.189* (1.753)	0.132 (1.240)	0.257** (2.151)	0.118 (0.975)	0.115 (0.943)	0.204 (1.504)	0.373** (2.106)	0.346** (2.143)	0.398** (2.523)	0.302* (1.942)
PTV4	0.146 (1.536)	0.184 (1.533)	0.183 (1.641)	0.265** (2.210)	0.249** (2.118)	0.22* (1.823)	0.133 (1.107)	0.171 (1.284)	0.054 (0.417)	0.155 (1.162)
PTV5	0.18 (1.638)	0.122 (1.230)	0.178 (1.614)	0.07 (0.639)	0.125 (1.058)	0.113 (1.052)	0.206* (1.823)	0.157 (1.284)	0.233* (1.914)	0.13 (1.131)
PTV6	0.316*** (3.157)	0.149 (1.556)	0.173 (1.550)	0.196 (1.625)	0.054 (0.537)	0.071 (0.704)	0.127 (1.118)	0.231* (1.842)	0.227* (1.783)	0.203* (1.715)
PTV7	0.333*** (3.334)	0.2* (1.900)	0.177* (1.731)	0.181* (1.750)	0.119 (1.029)	0.263** (1.983)	0.169 (1.487)	0.105 (0.931)	0.145 (1.316)	0.15 (1.274)
PTV8	0.398*** (3.787)	0.263*** (2.770)	0.191* (1.885)	0.265** (2.471)	0.188* (1.896)	0.155 (1.499)	0.169 (1.482)	0.168 (1.402)	0.208* (1.711)	0.243** (2.060)
PTV9	0.55*** (3.979)	0.479*** (3.895)	0.434*** (4.150)	0.285*** (2.772)	0.259** (2.535)	0.249*** (2.716)	0.201** (2.345)	0.202** (2.254)	0.231** (2.503)	0.223** (2.400)
High	0.324* (1.787)	0.545*** (4.133)	0.477*** (4.019)	0.484*** (4.148)	0.511*** (4.468)	0.491*** (4.346)	0.431*** (3.978)	0.436*** (4.397)	0.409*** (4.293)	0.351*** (3.823)
Low-High	0.018 (0.100)	0.507*** (2.646)	0.434** (2.339)	0.267 (1.518)	0.260 (1.561)	0.096 (0.586)	0.092 (0.537)	0.166 (1.025)	0.160 (0.998)	0.175 (1.076)
<i>Panel B. Expected utility value (EU) and PTV</i>										
EU	0.345*** (5.681)	0.0541** (2.063)	-0.0021 (-0.312)	-0.0008 (-0.856)	0.0000 (-0.23)	0.0000 (0.291)	0.0000 (1.209)	0.0000* (1.935)	0.0000 (0.917)	0.0000 (0.999)
PTV	-0.2494*** (-20.745)	-0.213*** (-22.849)	0.2067*** (23.081)	-0.2081*** (-23.133)	-0.207*** (-23.14)	-0.2056*** (-23.073)	-0.2057*** (-23.21)	0.2061*** (-23.107)	-0.2037*** (-22.926)	-0.2034*** (-22.899)
Note(s): The table summarizes the analysis with EUT as the variable of interest. Panel A illustrates the portfolio analysis and presents ten portfolios created using varying values of the θ parameter within the range of 0.5–10. The lowest EUT value was assigned to the first portfolio, and only relevant coefficients are displayed. Only the relevant coefficients are reported. Panel B reports the regression results with EUT as one of the explanatory variables. Further, *, ** and *** denote statistical significance at 10, 5 and 1 % levels										

From a practical perspective, my findings have implications for investors and policymakers. Recognizing the dependence pattern would help policymakers discern the market's direction and reduce the volatility resulting from limited rationality among investors. I contend the usefulness of implementing a trading strategy centered around the PTV effect, specifically during normal and high prices. I note the daily average and risk-adjusted return difference of at least 0.79% between portfolios with high and low PTV.

In conclusion, my research significantly contributes to the CC and behavioral finance literature by highlighting the important relationship between PTV and CC returns, enabling a better understanding of CC price dynamics. Modeling the relationship across the return quantiles provides more profound and comprehensive insights into the relationship. Future researchers can enhance my work by incorporating models that more accurately embody the representation and valuation stages of decision-making under risk. This will contribute to a better understanding of investors' decision-making processes, allowing for a more realistic analysis of their behaviors and choices. By addressing these aspects, future studies can advance the field and provide valuable insights into the intricacies of investment decision-making.

Notes

1. The influence of sentiments can be gauged from instances like- In January 2021, the price of Dogecoin galloped over 300 percent in 4 hours following a tweet by Elon Musk.
2. Sourced from www.coinmarketcap.com
3. Sourced from www.coinmarketcap.com
4. I also conduct the analysis using raw return, excess return to the risk-free rate, findings do not change and are provided in robustness section.
5. While removing such observations I also exclude the next 29 observations from the sample, to remove its impact on other variables (for most variables my construction period window is 30 days).
6. Table 2B in Appendix presents the definition of control variables used in the study
7. To address the impact of changing economic conditions over time, I introduce yearly time fixed effects into my analysis. This is because economic fundamentals generally do not undergo significant fluctuations within a single year. Additionally, comprehensive QR model is provided in Table B4 that includes all relevant variables suggested in the literature along with time dummy. However, in my primary analysis, I have constrained the number of control variables to address the issue of multicollinearity.
8. To account for temporal fluctuations, I include yearly time fixed effects in my analysis. However, I intentionally omit fixed individual effects to accurately distinguish between underperforming and overperforming CCs, as suggested by Barnes and Hughes (2002). I also examine the relation between PTV and CC returns by applying QR in the second stage of Fama and Macbeth (1973). Results are qualitatively similar.
9. Barber and Odean (2008) argue that retail investors are net buyers of attention-grabbing stocks and attribute it to their limited information processing abilities. Evidence of limited cognitive abilities is also provided by Grossman and Stiglitz (1980) and Tversky and Kahneman (1974). Thus, the study underlines the evidence of the attention-induced price pressure hypothesis.
10. This also indicates that existing CC persistence indicators like momentum cannot capture the predictive power of PTV. Nagel (2002) credit under-reaction as the most promising behavioral explanation of momentum. Barberis *et al.* (1998) attribute this under-reaction to self-attribution bias, whereas Hong and Stein (1999) assign it to the slow diffusion of information and bounded rationality.

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Appendix

The supplementary material for this article can be found online.

Behavioral
biases of CC
investors

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